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Numerical Stability of Functional MRI Connectivity Biomarkers

Haute École d'Ingénierie de Sion · Informatique et Systèmes de communication · Data engineering

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Context and Motivation

Functional Magnetic Resonance Imaging (fMRI) enables non-invasive observation of brain activity. A key application is Functional Connectivity (FC) analysis, which extracts matrices encoding undirected graphs of regional signal correlation. FC is promised to serve as biomarkers for conditions such as Major Depressive Disorder (MDD).

As pipelines become more computationally intensive, small perturbations from Operating System (OS) differences, hardware, and parallelization can propagate and introduce spurious variability and correlation into FC results.

Figure 1: Same program, different results across operating systems.

Methodology

Monte Carlo Arithmetic via Fuzzy introduces controlled random perturbations to every floating-point operation during preprocessing steps, simulating numerical variability across different computers.

The first two analyses reused the perturbed fMRIPrep 25.2.5's preprocessing outputs:

1. **Graph metrics**: the Numerical Population Variability Ratio (NPVR) was used to assess the stability of network summary measures on perturbed FC matrices across thresholds and brain regions (Alizadeh et al., 2026).
2. **Edge-wise FC**: NPVR was computed for each of the 4,950 edges, and the effect of global signal regression on numerical stability was quantified.

The third analysis perturbed a separate feature extraction pipeline:

3. **Principal Component Analysis (PCA) biomarkers**: the numerical stability of the PCA-based feature extraction method of Yamashita et al., 2026 was assessed by perturbing `np.corrcoef` with MCA and forcing the PCA to 32-bit inputs, on SRPB and BMB datasets.

Graph Metrics Reproduction

Reproduced on a different multi-site dataset provided by ATR (Kyoto) with more extensive confound regression (15 vs. 6 regressors). Graph metrics displayed broadly similar trends to the original study, but with consistently higher NPVR values across all metrics and thresholds.

The confound regression effect was considerably larger and more variable across thresholds than originally reported, with differences spanning a much wider range.

Finding: Graph metrics follow similar trends to the original study, but with consistently higher NPVR across metrics and thresholds. Using more confounds (15 vs. 6) further raises this variability and widens its spread.

Edge-wise FC Stability

Edge-wise NPVR ranged from 0.036 to 0.297 (mean 0.11), with clear spatial patterns of numerical sensitivity across the brain. Global signal regression halved variability across most of the 4,950 connections, with the reduction widespread rather than concentrated in specific regions.

Figure 2: Left: NPVR heatmap. Right: delta NPVR after removing global signal regression.

Figure 3: Left: edge-wise NPVR distribution. Right: delta NPVR distribution.

Finding: Edge-wise stability is uneven. Global signal regression reduces numerical variability across most connections.

PCA Biomarker Stability

Reproduced on complete SRPB and BMB datasets, both showed consistent MDD under-connectivity across prefrontal, motor, and subcortical networks, though regional dominance varied.

PCA biomarkers were identical across all perturbed runs (100 SRPB, 15 BMB), even with MCA-perturbed `np.corrcoef` and 32-bit inputs.

Figure 4: Identical SRPB biomarkers from a single run (left) and 100 perturbed runs (right). Red = over-connected, blue = under-connected in MDD.

Finding: PCA biomarkers are numerically stable: identical connections across all perturbed runs, with the same MDD under-connectivity pattern generalizing across datasets despite varying regional importance.

Key Takeaways

- **Graph metrics** reproduce similar trends but with higher NPVR than originally reported; more extensive confound regression (15 vs. 6 regressors) amplifies variability more than originally reported.
- **Edge-wise FC** shows uneven stability (NPVR 0.036–0.297, mean 0.11). Global signal regression halves variability across most connections, with widespread rather than region-specific reduction.
- **PCA biomarkers** are stable within each dataset (identical connections across all perturbed runs). The same MDD under-connectivity pattern generalizes across SRPB and BMB, but regional importance and specific connections vary.

Open questions: how do other PCA pipeline steps respond to perturbation, and what is each confound's precise effect on NPVR?